Grid-Forming Control of a PMSG Wind Turbine Using Rotor Kinetic-Energy Reserve for Inertia and Fast Frequency Support in Weak Grids

Step 1 Progress Report, Mid-Semester Evaluation

Wind Electrical Systems Design Project

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The problem: converter-interfaced wind gives no inertia, and grid-following control fails in weak grids

- Today's turbine is grid-following: a PLL measures the grid frequency and the converter injects MPPT power. The rotor's kinetic energy is hidden, so the turbine adds zero inertia.
- In a weak grid the PLL tracks a voltage that the converter itself disturbs, and it can lose synchronism.
- GRID-INDIA's discussion paper on grid-forming inverters (Feb 2026) recommends GFM capability but treats BESS and solar in detail and says little about wind.
- **This project: make the wind turbine itself grid-forming, with its reserve and inertia coming from the rotor, not from a battery.**

54 GW

wind installed in India by late 2025

68 events

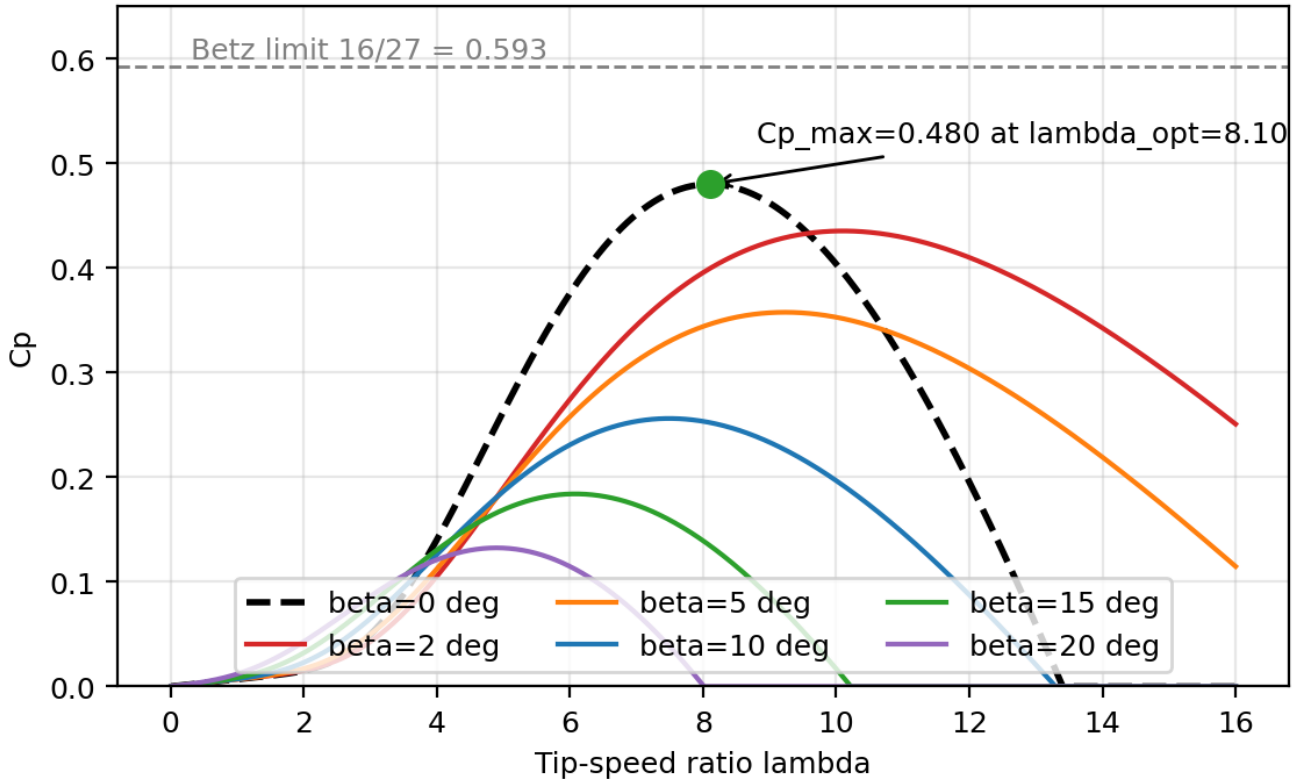
more than 1 GW of renewables lost in a single incident, Jan 2022 to Jul 2025 (Ember); PLL instability under weak-grid conditions, Rajasthan hubs

1 Apr 2027

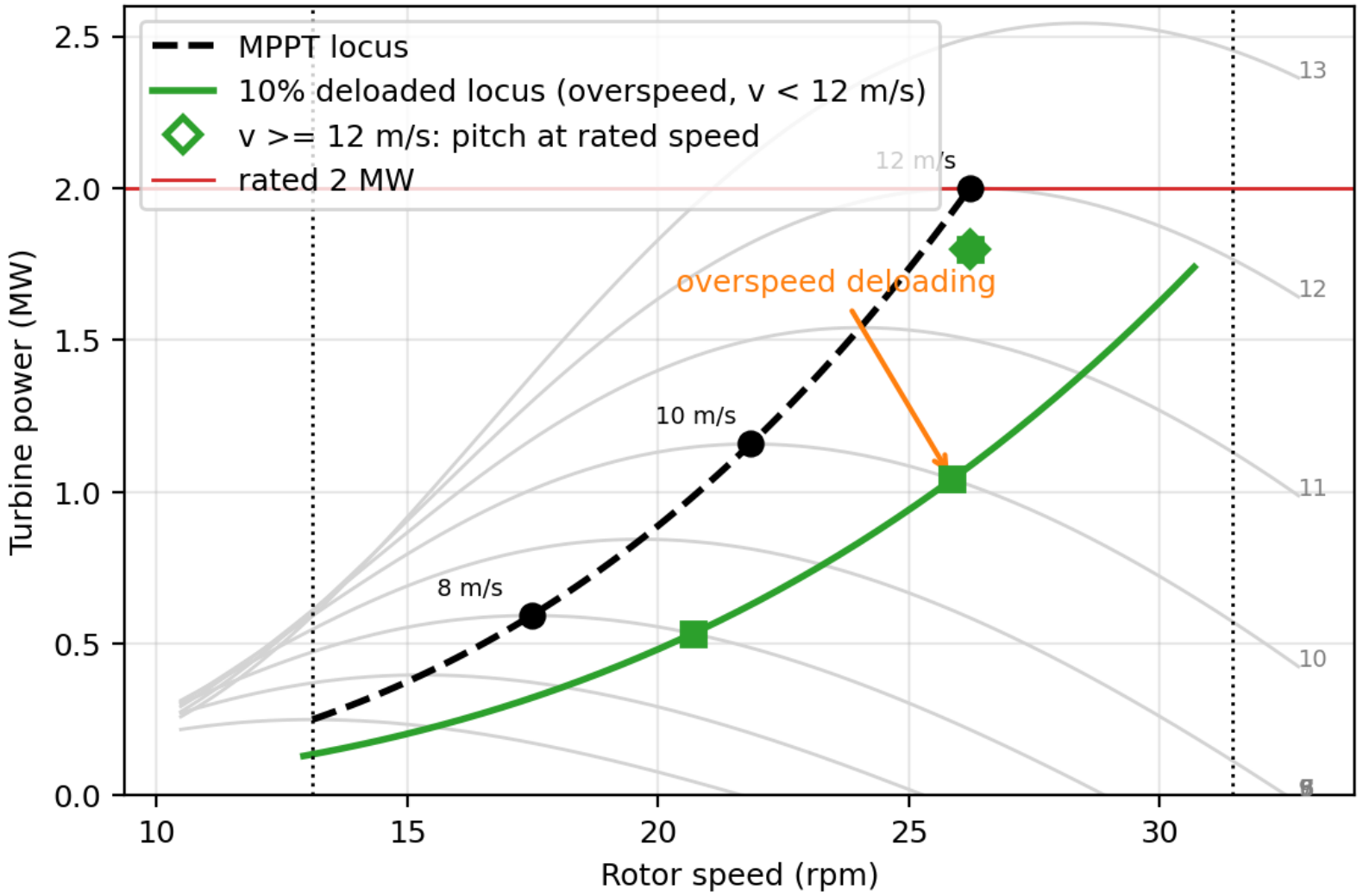
CEA technical-standards amendment 2026 takes effect

Link to the course: Betz, the Cp-lambda curve and the Cp-optimal line

- Power in the wind: $P = \frac{1}{2} \rho A v^3$. Betz limit 16/27 = 0.593.
- Heier model $Cp(\lambda, \beta)$ for the 2 MW turbine: Cp,max = 0.480 at λ_opt = 8.10, 81 percent of Betz (drag, tip loss, wake rotation).
- MPPT = staying on the peak, the Cp-optimal line: $P = K_opt\ \omega^3$ with K_opt = 96.6 kW/(rad/s)³.
- Course control strategies used here: pitch control (mechanical power) and controlling the electrical power through the power-electronic interface.
- Turbine: R = 35.4 m, 2 MW at 12 m/s and 26.2 rpm, H = 3 s, speed limits 0.5 to 1.2 pu.



Key idea: operate off the Cp-optimal line, on the overspeed side



At 9 m/s

MPPT: 844 kW at 19.7 rpm

Deloaded 10 percent: 759 kW at 23.3 rpm, pitch 0 deg

Reserve: 84 kW per turbine

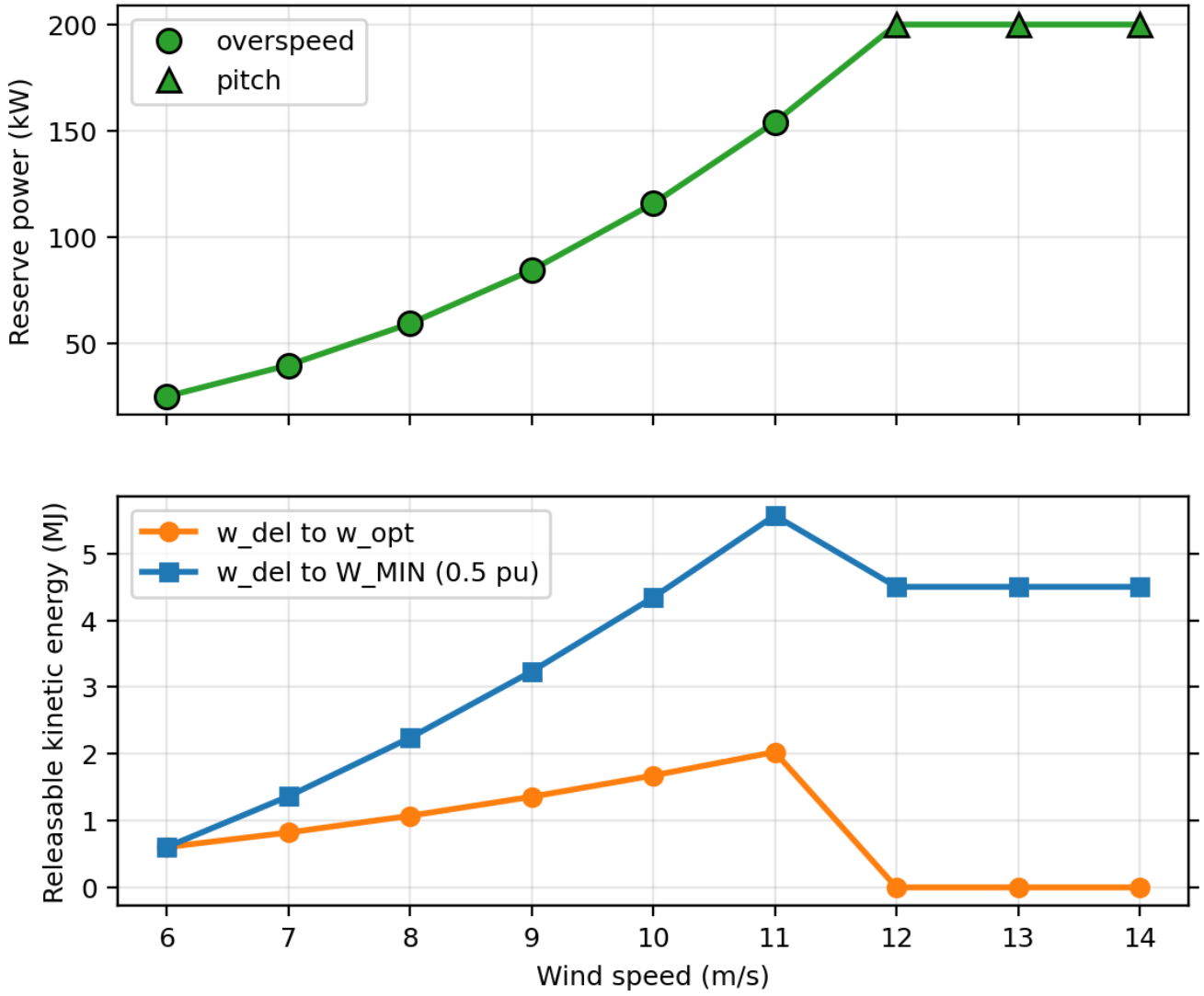
Why the overspeed side

When the converter draws more power the rotor slows toward the peak, so Cp RISES and aerodynamic power increases while the grid needs it.

Kinetic energy and reserve are released together: no secondary dip.

Above 12 m/s the 1.2 pu speed limit binds and pitch does the deloading (1.18 deg at 12 m/s, 8.5 deg at 14 m/s).

What the reserve buys: power headroom plus stored kinetic energy



25 → 200 kW

reserve per turbine, 6 m/s to rated (84 kW at 9 m/s)

1.36 MJ / 3.23 MJ

kinetic energy releasable at 9 m/s: to MPPT speed / to 0.5 pu (H_eq 0.68 s / 1.62 s)

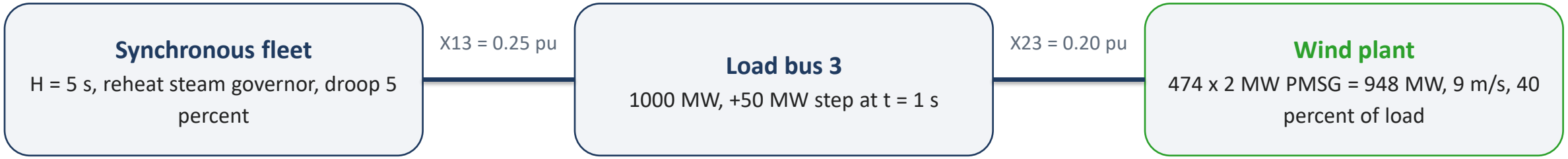
4,725 → 4,253 MWh/yr

annual energy, MPPT vs 10 percent deloaded (CF 27.0 → 24.3 percent), Weibull k = 2, c = 7.5 m/s

INR 16.5 lakh/yr

upper-bound cost at 3.5 INR/kWh; reserve is needed only in low-inertia hours, and it replaces a battery

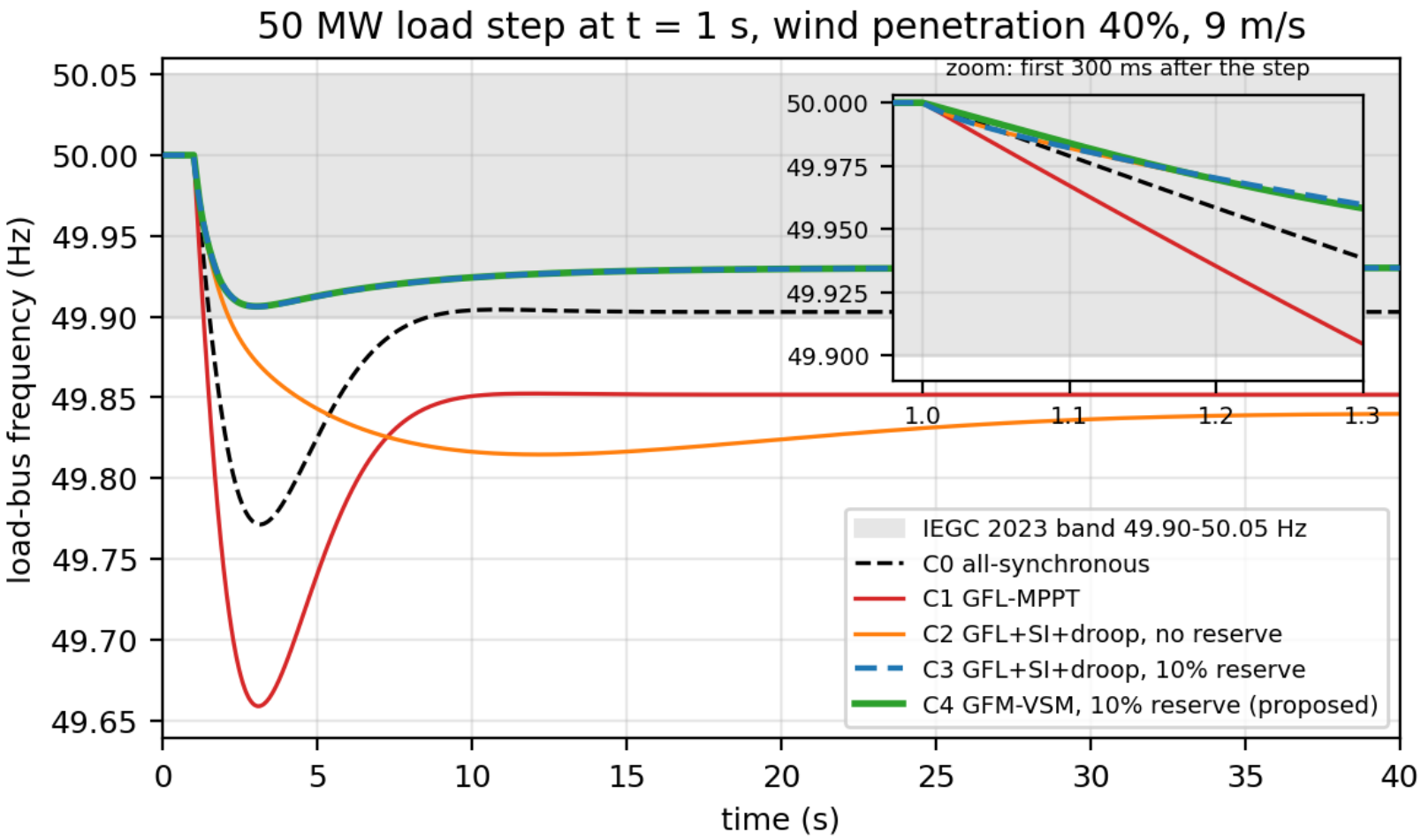
Test system and the five cases



Base 1000 MVA, 50 Hz. Load self-regulation D = 1. Controller gains identical in C2, C3, C4: virtual inertia H_v = 5 s, droop 5 percent on the wind base.

Case	Wind-plant control	Reserve	What it represents
C0	no wind, all synchronous	-	reference grid
C1	grid-following, MPPT	none	today's standard practice
C2	grid-following + synthetic inertia + droop	none	what vendors offer now; support is borrowed from the rotor
C3	grid-following + synthetic inertia + droop	10 percent overspeed	same controller as C2, with the reserve
C4	**grid-forming virtual synchronous machine (proposed)**	**10 percent overspeed**	**voltage source: inertia and droop with no frequency measurement**

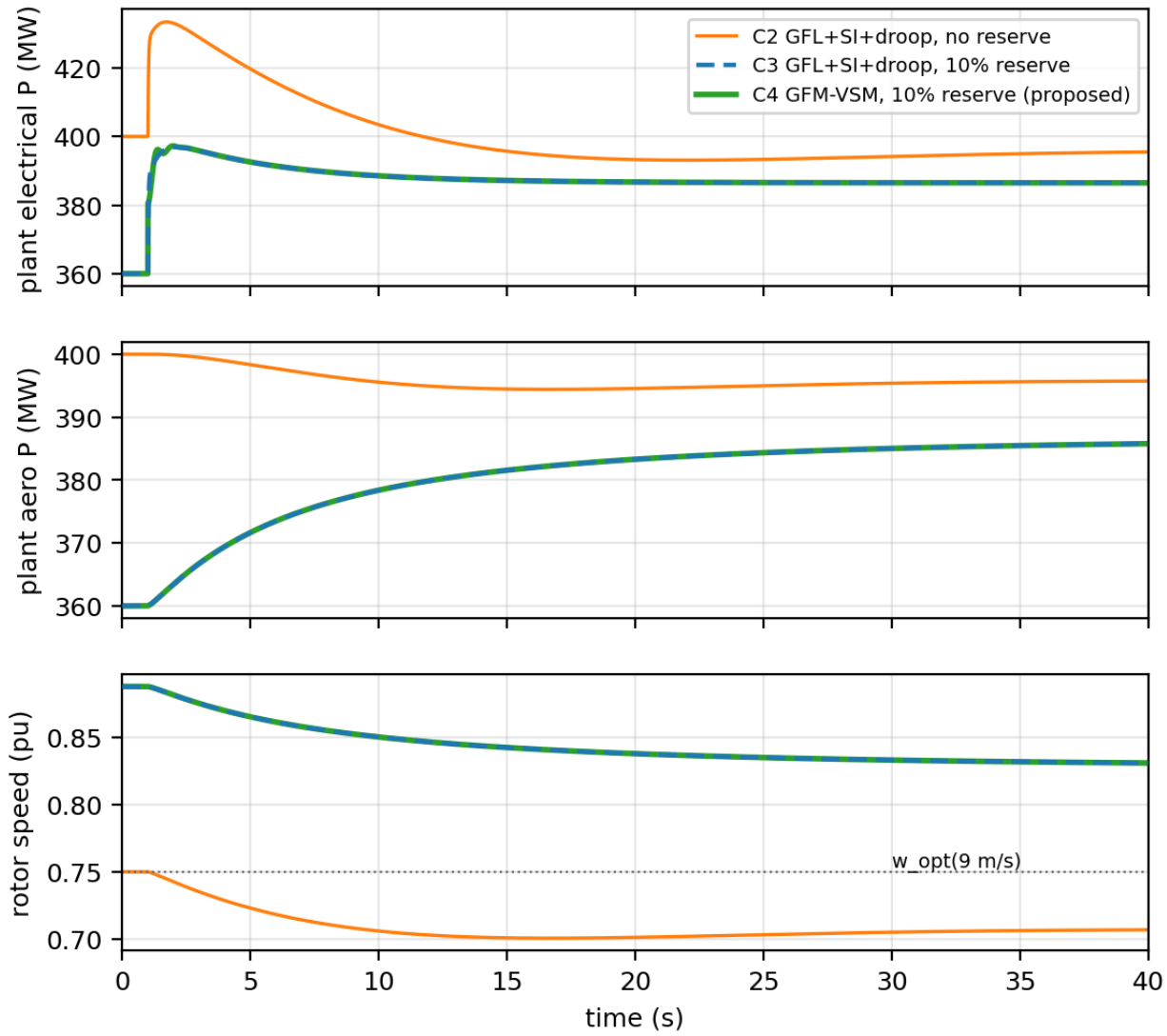
Headline: grid-forming with the reserve keeps the nadir inside the IEGC band



Case	Nadir (Hz)	RoCoF at step
C0 all-sync	49.771	0.213 Hz/s
C1 GFL MPPT	49.659	0.332 Hz/s
C2 GFL + SI	49.815 (11 s)	0.332 Hz/s
C3 GFL + reserve	49.906	0.332 Hz/s
C4 GFM + reserve	**49.906**	**0.162 Hz/s**

System inertia H_sys: 3.76 s (C1) → 8.51 s (C4)
The reserve sets the nadir: C3 and C4 tie at 49.906 Hz.
Grid-forming halves the instantaneous RoCoF because it responds as a voltage source with no frequency measurement.

Why it works: the wind plant during the event



Electrical power (top)

C2 spikes to 433 MW and decays: it only borrows. C4 steps from 360 to 397 MW and holds about 387 MW.

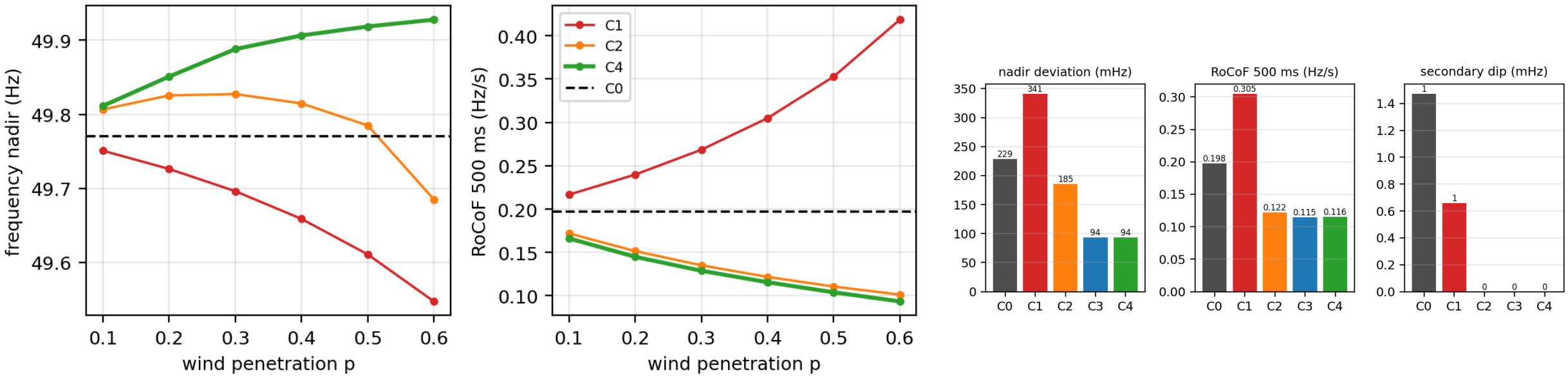
Aerodynamic power (middle)

C2 falls to 394 MW as the rotor leaves the optimum. C4 rises from 360 to 386 MW as the rotor slows toward the peak.

Rotor speed (bottom)

C2 drops below the MPPT speed, 0.750 to 0.701 pu. C4 stays above it, 0.888 to 0.831 pu: the extra 37 MW is reserve plus kinetic energy.

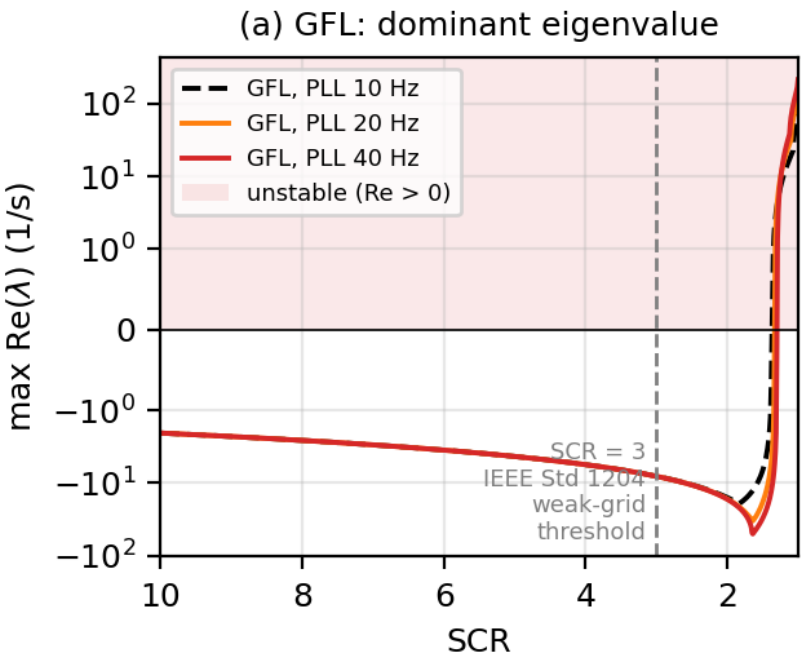
Metrics by case, and what happens as wind share grows

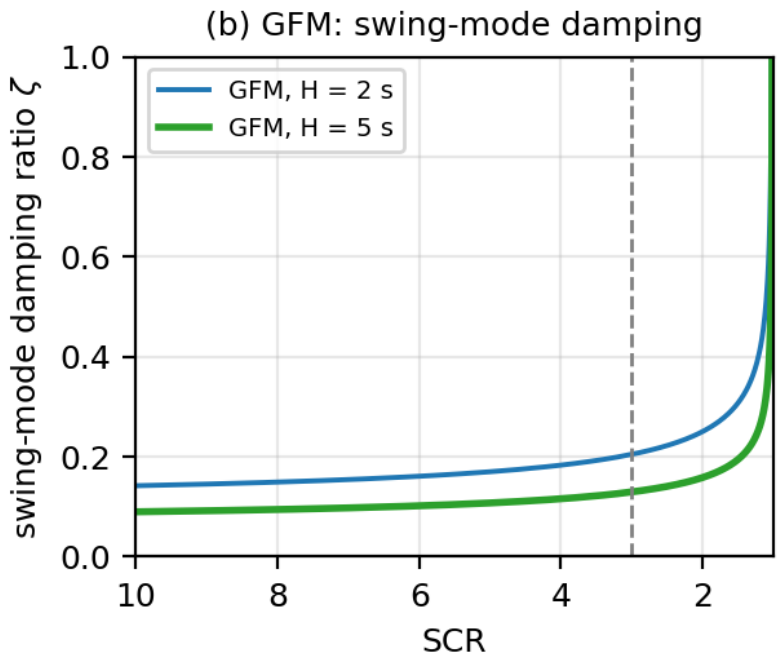


Left (B4): nadir and 500 ms RoCoF versus wind share. Right (B3): nadir deviation, 500 ms RoCoF and secondary dip per case at 40 percent wind.

10 to 60 percent wind: grid-following nadir worsens 49.751 → 49.547 Hz; grid-forming nadir improves 49.811 → 49.928 Hz. Every grid-forming turbine adds inertia and reserve, so more wind strengthens the grid.

Weak-grid stability: grid-following fails where grid-forming gets better





SCR 1.30 to 1.37

grid-following loses stability (40 / 20 / 10 Hz PLL); IEEE 1204 weak-grid threshold is SCR 3

stable to SCR 1.0

grid-forming VSM stays stable to the static limit; damping rises as the grid weakens

ζ 0.25 → 0.60

damping of the proposed plant as its connection weakens from SCR 3 to 1 (nadir unchanged)

Duality (Li, Gu and Green, 2022): grid-following likes a stiff grid, grid-forming likes a weak one. Weak Indian RE hubs are exactly where grid-forming wind belongs.

Quasi-static model: PLL bandwidth shifts the critical SCR only slightly here; the fast-PLL penalty needs the electromagnetic network dynamics planned for Step 2.

Verification: 16 checks against hand-derivable results, all pass

Gate	What it checks	Value	Result
GA1	Cp,max below the Betz limit	0.480 < 0.593	PASS
GA2	λ_opt and Cp,max in the expected range for the Heier model	8.10, 0.480	PASS
GA3	MPPT power at 12 m/s equals the 2 MW rating	2.000 MW	PASS
GA4	Deloaded operating point reproduces 0.9 P_avail at every wind speed	max error 6e-14	PASS
GA5	Deloaded speed above MPPT speed in every overspeed case	yes, 6 to 11 m/s	PASS
GA6	Annual-energy loss of the deloaded case equals 10 percent	10.000 percent	PASS
GB1	Exact pre-event steady state (no drift before the step)	|Δf| = 0	PASS
GB2	Initial COI RoCoF equals ΔP f0 / (2 H_sys), cases C0, C1, C4	errors 0.19, 0.24, 0.93 percent	PASS
GB3	C1 steady-state Δf equals the droop and load-damping formula	error 1e-10	PASS
GB4	Rotor energy balance: ∫(P_aero − P_e) dt equals ΔE_kinetic	error 1e-11	PASS
GB5	Proposed case keeps the rotor above the MPPT speed	min 0.831 pu (limit 0.740)	PASS
GB6	Ordering: C4 RoCoF below C1, C4 nadir above C1 and C2	0.116 < 0.305 Hz/s	PASS
GC1	GFL static limit without voltage support at SCR = 2 P0	within 2 percent	PASS
GC2	GFM swing mode matches closed-form ω_n and ζ at SCR 5	within 2 percent	PASS
GC3	Every GFL variant stable at SCR 10	yes	PASS
GC4	GFL margin falls as SCR falls; GFM damping falls as SCR rises	yes	PASS

Steps 2 and 3

Step 2: detailed converter model

- PMSG dq model with machine-side and grid-side converters and the DC link
- Dual-port grid-forming control (Lyu and Gross, 2024)
- Current limiting and fault ride-through
- Realistic frequency-measurement windows for the grid-following comparison
- Weak-grid study with electromagnetic network dynamics (fast-PLL penalty)

Step 3: system test and final report

- IEEE 9-bus test system with the detailed plant
- Turbulent wind input and wind-speed sensitivity
- Check against Indian grid-code requirements (IEGC 2023, CEA 2026)
- Design recommendation for wind plants in weak Indian grids
- Final report

Step 1 delivered: aerodynamic design, frequency-support simulation, weak-grid stability study, 16 verified checks. Grid-forming wind with a rotor reserve keeps the nadir inside the IEGC band and stays stable where grid-following fails.

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